

2 Equations, formulae and identities

	Students should be taught to:	Notes
2.1 Use of symbols	A understand that symbols may be used to represent numbers in equations or variables in expressions and formulae	
	B understand that algebraic expressions follow the generalised rules of arithmetic	
	C use index notation for positive and negative integer powers (including zero)	$a \times a \times a = a^3$ $a^{-5} = \frac{1}{a^5}$; $a^0 = 1$
	D use index laws in simple cases	$x^m \times x^n = x^{m+n}$ $x^m \div x^n = x^{m-n}$ $(x^m)^n = x^{mn}$
2.2 Algebraic manipulation	A evaluate expressions by substituting numerical values for letters	
	B collect like terms	
	C multiply a single term over a bracket	$3x(2x + 5)$
	D take out common factors	Factorise fully $8xy + 12y^2$
	E expand the product of two simple linear expressions	Expand and simplify $(x + 8)(x - 5)$
	F understand the concept of a quadratic expression and be able to factorise such expressions (limited to $x^2 + bx + c$)	Factorise $x^2 + 10x + 24$
2.3 Expressions and formulae	A understand that a letter may represent an unknown number or a variable	
	B use correct notational conventions for algebraic expressions and formulae	
	C substitute positive and negative integers, decimals and fractions for words and letters in expressions and formulae	Evaluate $2x - 3y$ when $x = 4$ and $y = -5$
	D use formulae from mathematics and other real-life contexts expressed initially in words or diagrammatic form and convert to letters and symbols	
	E derive a formula or expression	
	F change the subject of a formula where the	Make r the subject of

	Students should be taught to:	Notes
2.4 Linear equations	A solve linear equations, with integer or fractional coefficients, in one unknown in which the unknown appears on either side or both sides of the equation	$5x + 8 = 12$ $7(x + 3) = 5x - 8$ $\frac{4x + 5}{2} = 3$
	B set up simple linear equations from given data	The three angles of a triangle are a° , $(a + 10)^\circ$, $(a + 20)^\circ$. Find the value of a
2.5 Proportion	Higher Tier only	
2.6 Simultaneous linear equations	A calculate the exact solution of two simultaneous equations in two unknowns	$x + y = 14, x - y = 2$ $2a + 5b = 12,$ $3a + b = 5$
2.7 Quadratic equations	A solve quadratic equations by factorisation (limited to $x^2 + bx + c = 0$)	Solve $x^2 + x - 30 = 0$
2.8 Inequalities	A understand and use the symbols $>, <, \geq$ and $\leq$	To include double-ended inequalities e.g. $1 < x \leq 5$
	B understand and use the convention for open and closed intervals on a number line	
	C solve simple linear inequalities in one variable and represent the solution set on a number line	$3x - 2 < 10$, so $x < 4$ $7 - x \leq 5$, so $x \geq 2$ $3 < x + 2 \leq 5$ so $1 < x \leq 3$
	D represent simple linear inequalities on rectangular Cartesian graphs	Shade the region defined by the inequalities $x \geq 0$, $y \geq 1$, $x + y \leq 5$
	E identify regions on rectangular Cartesian graphs defined by simple linear inequalities	Conventions for the inclusion of boundaries are not required